

## Events

An event is any action that a program can detect, such as a mouse click or a keystroke. The section of code that reacts to an event is called an “event handler”. Event handlers are used a lot in JavaScript and can trigger many different functions, making web pages fun and interactive.

### 1 Type the code

In this example, an event (clicking a button) triggers a simple function (a tongue-twister appears). Type the code in an IDLE code window and save the file with a “.html” ending.

```
<button onclick="tonguetwist()">Say this!</button>
<script>
  function tonguetwist()
  {
    document.write("She sells seashells");
  }
</script>
```

Annotations:

- Name of the function (points to `tonguetwist()`)
- HTML code links the function to a button (points to `onclick="tonguetwist()"`)
- Curly brackets surround a block of code—they work in a similar way to indents in Python (points to the curly braces around the function body)
- JavaScript code defines the function (points to the function definition)

### 2 Run the program

Double-click the file to launch the program in a browser window.



## Loops in JavaScript

A loop is a section of code that repeats. Using loops is much quicker and easier than typing out the same line of code over and over again.

### 1 Loop code

Like Python, JavaScript uses “for” to set up a loop. The repeated lines of code are enclosed in curly brackets. This loop creates a simple counter that increases by one each time it repeats.

```
<script>
for (var x=0; x<6; x++)
{
  document.write("Loop count: "+x+"<br>");
}
</script>
```

Annotations:

- The “<script>” tag introduces the JavaScript code (points to `<script>`)
- This line creates a counter called “x” with a starting value of 0, rising by 1 in each repeat (points to `x=0; x<6; x++`)
- This line tells the computer to write the text “Loop count:” followed by counter number (points to `document.write("Loop count: "+x+"<br>");`)

### 2 Loop output

Save the code as a “.html” file and run it. The loop keeps repeating as long as “x” is less than 6 (“x<6” in the code). To increase the number of repeats, use a higher number after the “<” symbol.

